

MT514

Three Phase Prepaid Meter with CPU Card

1. General

MT514 is a new type of 3 phases 4 wires CPU card prepaid electronic energy meter, which developed and produced by Iskraemeco for residential, commercial and industrial customers. It is designed, manufactured, test under the requirements of IEC62053-21, IEC62053-22, IEC62053-23, IEC62056-21(IEC61107) and other relevant international standards.

Main features :

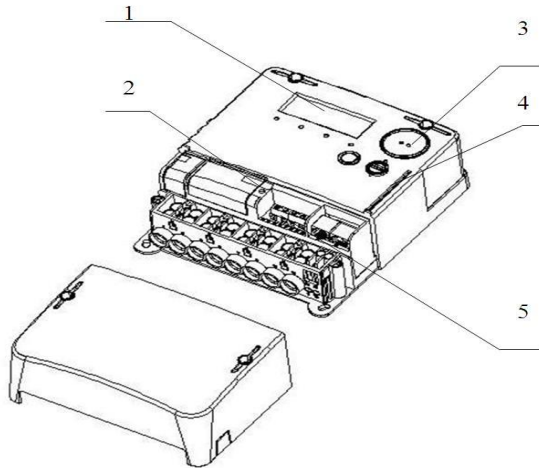
- ☆ Active, reactive energy measurement
- ☆ Adopt logic encryption card realizing two-way data transmission, support remote charge.
- ☆ Support step tariff, maximum 9 steps 10 tariff or TOU tariff maximum 8 tariffs
- ☆ Offset when consumption pass several grades
- ☆ Support prepayment mode and postpaid mode
- ☆ Different time different MPL(maximum power limit)
- ☆ Over MPL alarm and trip. The MPL is programmable.
- ☆ Low credit alarm, low credit alarm limit is programmable
- ☆ When the customer card missed, the system will reissue the card for customer.
- ☆ Terminal cover open detection, meter cover open detection
- ☆ Relay open when power failure, relay close when power resume(optional)
- ☆ Measure voltage, current, power, angle, frequency of each phase
- ☆ LCD display with backlight
- ☆ Measure current of neutral line
- ☆ Measure maximum demand
- ☆ Prevent tamper and event record
- ☆ Optical communication, the protocol is IEC61107
- ☆ GPRS communication (optional)
- ☆ RS485 communication(optional)
- ☆ Load profile function(optional)

2. Spec and technical parameters

Item	Technical spec (refer to the parameters on the nameplate)
Rated voltage	3×220V/380V
Range of operation voltage	60%Un~ 120%Un
Rated current	3×1(10)A
Pulse constant	Active energy: 8000imp/kWh; Reactive energy: 8000imp/kvarh
Starting current	0.1% Ib
Accuracy class	Active energy: Class0.5 ; Reactive energy: 1.0
Reference frequency	50Hz
Power consumption	≤2W, 6VA
Temperature	-25℃~+55℃
Temperature limit	-40℃~+70℃
Humidity	<95%(Free of condensation)
Outline dimension	L×W×H=263mm×176mm×72mm
Weight	1.6kg
Protection class	IP54
Impulse withstand voltage	6kV
AC voltage	4kV
Electrical fast transient	4kV
Pulse parameters	Pulse width:(35±16)ms, Current≤10mA, Voltage: DC(5~24)V(External power source)

3. Operation principal

3.1 Meter outlook:



3.1.1 LCD

3.1.2 Terminal cover open detector

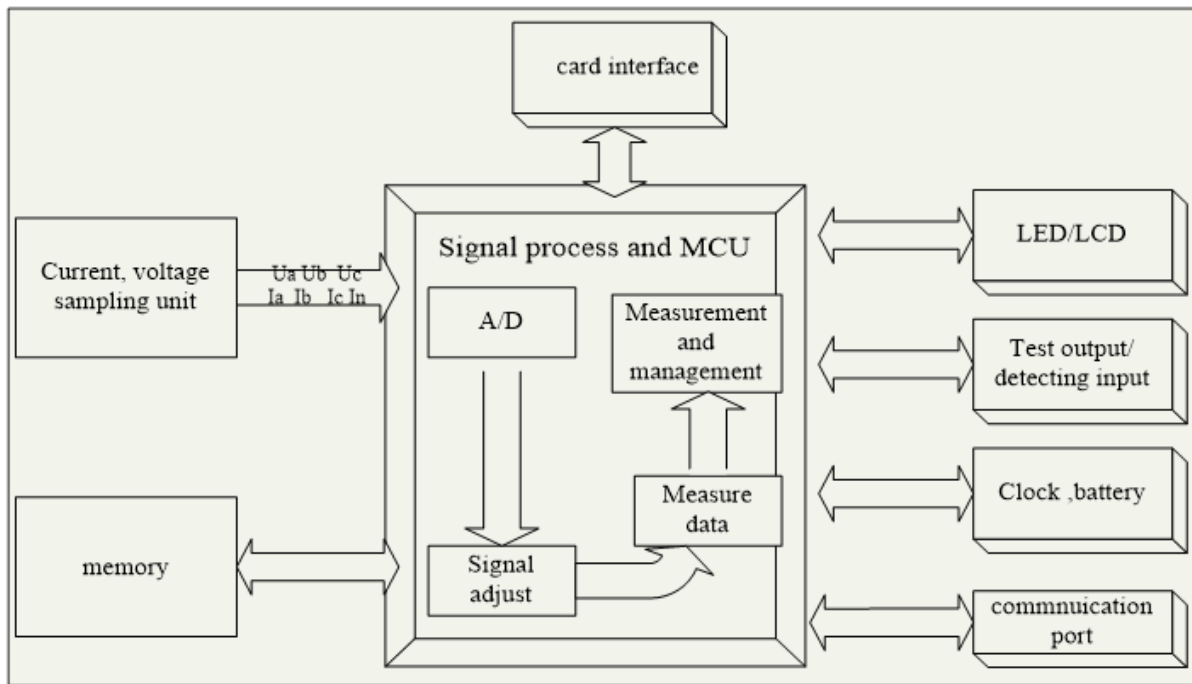
3.1.3 Optical communication port

3.1.4 CPU Card slot

3.1.5 Auxiliary terminal

3.2 Operation principal

Sampling current and voltage signal by shunt resistor and divider resistor net, it uses single Chip microcomputer to complete measurement, management, communication, data saving, driving liquid crystal display etc. Measurement chips work with MCU to convert the sampled analog signal into digital signal. Then take an integral operation to acquire the accurate measurements. Per processing, can provide or exchange information with external interface at any time.



The diagram of operation principal

4. Main functions

4.1 Measure active and reactive energy

Total energy= import energy + export energy

4.2 Measure total active energy

There are 2 methods about the measure total active energy, we choose 1 when calibration.

1. Total active energy is the sum of absolute of 3 phases, $P^+ = |P_A| + |P_B| + |P_C|$;
 $Q^+ = |Q_A| + |Q_B| + |Q_C|$
 2. Total active energy is the sum of algebra of 3 phases, $P^+ = P_A + P_B + P_C$;
 $Q^+ = Q_A + Q_B + Q_C$.
- 1 is the default one.

4.3 Measure average power

Measure each phase and total phase average power. The average power is the mean of each minute instantaneous power, can be used for load controlling. The minimum unit for active power is 0.0001kW.

4.4 Relay open automatically when remain credit use out.

Relay open when remaining credit to be 0 (not weekend not friendly time), user should purchase the energy to resume the power supply. (When remaining credit is bigger than 0, relay close automatically)

4.5 Increment function

Upon acceptance of a valid CPU Card, meter shall credit the exact amount contained in the card to the appropriate register in the meter and will increment the register by this amount

4.6 Step tariff

When the consumption energy is less than step 1, using tariff 1

When the consumption energy is more than step 1 less than step 2, using tariff 2

.....

We can make offset when pass steps. Eg: If the offset 1 is 50 currency, step 1 is 100kWh, when the meter consumption of current month is 101kWh, the meter will reduce this 50 currency.

4.7 Billing function

Bill date can be set at any date between 1st and 28th. Default billing time on each month is the 1st day at 0 o'clock, can store last 12 months data, billing data details as below:

1. Cumulative total consumption energy (import/export, active/reactive)
2. Monthly credit consumption (energy consumption in one month)
3. Monthly energy consumption (energy consumption in one month)
4. Active max demand and happen time

4.8 Low credit alarm

This meter have 3 levels of remaining energy alarm limit to remind user purchase energy:

When remaining energy is more than alarm limit1, LED alarm indicator will be green and long lighted.

When remaining energy is less or equal to alarm limit 1, and more than alarm limit 2, LED alarm indicator will be green and glitter.

When remaining energy is less or equal to alarm limit 2, and more than alarm limit 3, LED alarm indicator will be red and long lighted, the buzzer will alarm, press button or inset CPU Card the buzzer alarm will suspended, the time of suspended can be set.

When remaining energy is less or equal to alarm limit 3, LED alarm indicator will be red and glitter. At the same time whether the relay open depend on the parameter trip limit, if trip, press button or inset CPU Card the relay will close.

When remaining energy run out, relay open. User should purchase the energy (remaining energy is bigger than 0) to close the relay.

4.9 Overload alarm and trip

MPL can be set to different value in different time and there is switch times one day of MPL.

The meter can detect over MPL by the average current of 3phase or any one phase, this controlled by meter mode word 7 bit 1.

Relay open automatically when the average current of 3 phase current or any one phase exceeded the max power limit, at the same time, the corresponding LCD alarm symbol will be flashed. According to the binary code on LCD, we can track out the alarm reason.

Over MPL for 5 minutes, the relay will disconnect automatically for 5 minutes, then the relay will connect automatically. When the relay disconnect for 3 times, the meter disconnect & never connect again except in case of using maintenance cards.

When the MPL is 0 or more than 200A, in order to protect the equipment, the meter will using 200A as limit.

4.10 MPPUL alarm

Meter measure each phase average power. If the maximum phase deducts the minimum one exceeded the power unbalance limit, LCD alarm symbol will be flashed. According to the binary code on LCD, we can track out the alarm reason. Alarm will disappear when phase power unbalance value within the limit. Phase power unbalance limit is programmable.

4.11 Relay error alarm and control relay

Meter monitor relay work status continuously, to confirm if the relay worked as per MCU order. If find some problem on relay, LCD alarm symbol will be

flashed. According to the binary code on LCD, we can track out the alarm reason. If no problem, no alarm symbol flash.

Meter have 2 methods to monitor relay status. One is hardware detection, the meter will detect the voltage of phase A output (3phase with voltage). When meter detected voltage after relay disconnect or meter didn't detect voltage after relay connect, meter will alarm relay error. Another is current detection. Only when meter detected current (any phase current more than 0.04A last 5 seconds) after relay disconnect. The 2 methods controlled by meter mode 7 bit 3.

Relay can be controlled by CPU Card or remote command, at this moment, the relay cannot be controlled by logic.

4.12 Open terminal cover and open meter cover

Unauthorized terminal cover will trigger an event log be recorded in non-volatile memory with date and time stamp. At the same time, the alarm symbol will be flashed on LCD and the load will disconnected immediately. According to the binary code on LCD, we can track out the alarm reason.

Unauthorized terminal cover will trigger an event log, no matter of meter at power on or power off state.

When the terminal cover opened, the LCD will display check, the code will display -01 or -02, -01 means the terminal cover opened before power off, -02 means terminal cover opened after power off.

Before open account, the meter will not detect the terminal cover open.

4.13 Voltage phase disorder alarm

When the voltage phase disorder, the LCD will display 'Check' the code display -03, L1 L2 L3 will be flash, the alarm LED will be light, if the disorder lasts 10 seconds, meter will record the event, the relay will open. Only the voltage more than 80V, the meter will detect the disorder.

When the disorder recovered for 10 seconds, the relay will close.

There is a mode word to control whether the relay trip when voltage phase disorder happen.

4.14 Reverse alarm

When one phase power is reverse (voltage > 20V, current > 0.004Ib), the LCD will display 'Check' the code display -10, the alarm LED will be light. If reverse lasts 10 seconds, meter will record the event, the relay will open.

When the disorder recovered for 10 seconds, the relay will close.

There is a mode word to control whether the relay trip when reverse happen.

4.15 Phase loss alarm

When the voltage and current of one phase is 0, the meter detect the phase loss, LCD will display 'Check', the code is -09, the LED will be light. If the event lasts 10 seconds, meter will record the event, the relay will open.

When the disorder recovered for 10 seconds, the relay will close.

There is a mode word to control whether the relay trip when reverse happen

4.16 Magnetic disturbance alarm (optional, according to contract)

This function is optional. When meter detect the magnetic disturbance lasts 5 seconds, LCD will display 'check', the code is -08, the LED will be light.

4.17 Over voltage alarm

When one phase voltage is over the limit, the LCD will display 'check', the code is -07, the LED will be light.

4.18 power off trip

When power off, the meter will trip. When power on, if there is no trip reason, the relay will close.

4.19 Power off display

When power off and there is external battery, pressing the display button, the LCD will display the following items one cycle: meter number, remain credit, total consumption active energy.

If the terminal cover be well placed before power off, open terminal cover when power off, the LCD will display tEr-open to indicate the terminal cover is open the meter will record the terminal cover open happen time.

If the meter cover be well placed before power off, open meter cover when power off, the LCD will display toP-open to indicate the terminal cover is open the meter will record the meter cover open happen time.

4.20 Instantaneous data measurement

The meter can measure voltage, current, power and display them.

4.21 missing neutral

Real-time measurement the current of neutral line, display on the LCD.

There are 2 methods to detect missing neutral. This 2 methods are controlled by the mode 7 bit 4.

1. At least one phase current is more than 5A, the minimum current less than the biggest phase 70%, the current of null line is less than 4.5A, and lasts 5 seconds, the meter will judge the missing neutral.

The LCD will display ‘check’, the code is -06.

2. Meter will compare the calculated neutral current and measure neutral current. Only when calculate neutral current or measure neutral current more than 2A. The measure neutral current is smaller than 10% calculate neutral current and last for 5 seconds, the meter will record this event as missing neutral the measure neutral current is smaller than 90% the measure neutral current the meter will judge part missing neutral.

4.22 Max demand record

This meter can calculate and store the forward active maximum demand (including power demand and current demand) by each billing intervals, as well as the max demand occurrence time. They will display on the LCD.

4.23 Remaining day’s alarm

According to current remaining credit and last month average daily consumption, we can get to know the days meter can last for. If the remaining days are less than the limit, alarm triggered. Remaining days limit is programmable.

If last month average daily consumption is 0, 5kWh be default as last month average daily consumption for remaining usage days calculation

4.24 Modify date and time by CPU Card

The data and time of this meter can be modified by CPU Card.

4.25 Friendly day, friendly time and weekend.

When the remaining credit use out and reach the overdraft limit, the relay will not disconnect if it is friendly day, friendly time, weekend until the remain credit reach to friendly limit.

The maximum friendly day is 208, format is year month day. If set 000501, means 1st May is friendly day each year.

Friendly time can be programmable.

Weekend can be programmable.

When remaining credit is 0 during friendly day, friendly time or weekend, the meter will not trip. But this work time have a limit time.

4.26 If one phase with current without voltage last for 3 seconds

The meter will calculate the energy using rated voltage and PF 1.0. At the same time the meter will alarm and record this event.

4.27 Special radio fees and special tariff

1. Special fee that is equal to .002 EGP for each KWH and up to 45 KWH (.09 EGP).

It is preferable to have the rate .002 and maximum limit 45KWH programmable.

One month consumption, special tariff energy 1(can be set), deduct special tariff 1(can be set).

E.g: Energy 1=10kWh, tariff 1=0.006 EGP

One month consumption, special tariff energy 2(can be set), deduct special tariff 2(can be set).

E.g: Energy2=10kWh, tariff2=0.003EGP

4. If the meter consumption is zero during a month (no consumption) the meter should deduct 3 EGP (can be set) with billing reset (if billing reset comes and no changes between last month cumulative energy and this month cumulative energy).

4.28 prepayment mode and credit mode (optional)

There is a mode word to control whether the meter can be switch between prepayment mode and credit mode.

When the control cable detect 220V, AC, the meter will operate as a normal 3ph meters, when not detect, the meter will decrease the credit.

4.29 Load profile (optional)

The meter has an individual flash memory chip.

Record interval can be programmed.

4.30 TOU tariff

There are 2 kids TOU parameters, one is current used, the other is standby. The switch time can be programmed for TOU time interval and TOU tariff.

14 season tables, define month day corresponding week table.

8 week tables, define Monday to Sunday corresponding daily table.

8 daily tables, define hour minutes corresponding TOU tariff No.

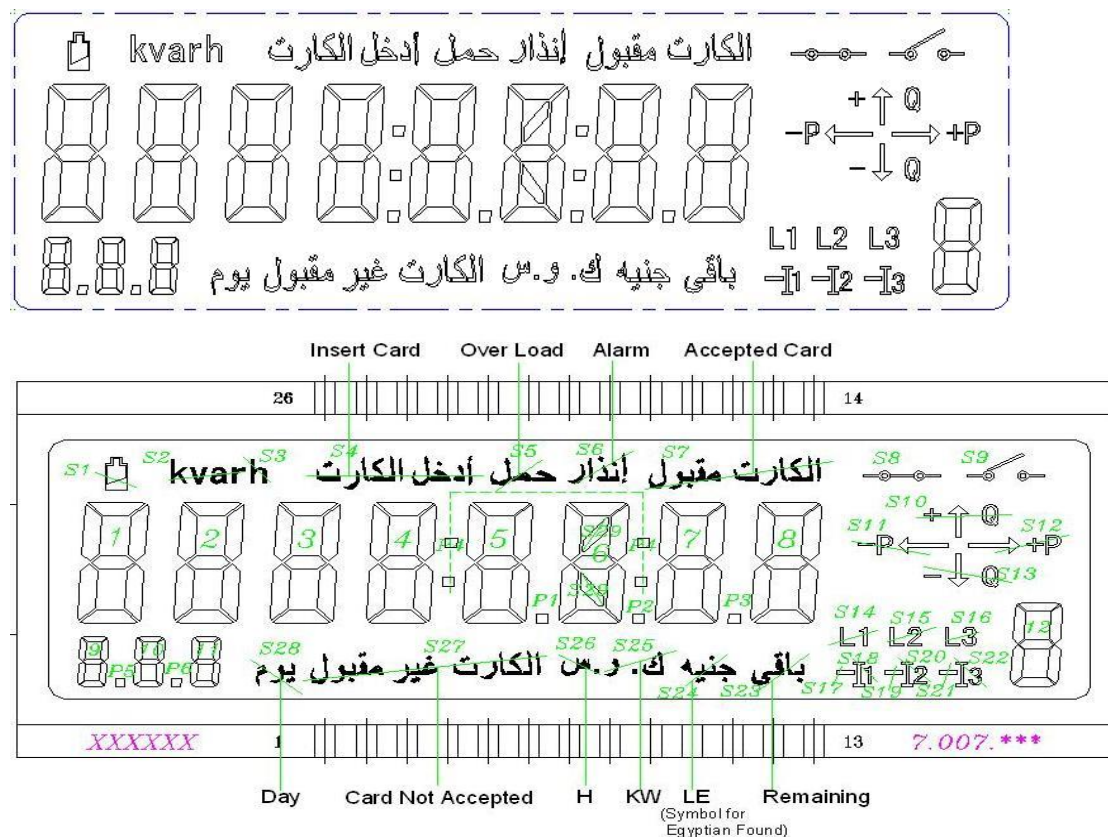
14 time interval each day.

8 TOU tariffs

The meter support 208 holidays and can choose any daily table for any holiday.

5. LCD display and LED indicator

5.1 Full-screen display



5.2 LCD displays instruction

: External battery low voltage

kvarh: Measurement unit

أدخل الكارت : Inset card indication

حمل : Current over Maximum current limit

إنذار : Alarm indication

الكارت مقبول : Meter deal with the card success

الكارت غير مقبول : Meter deal with the card fail

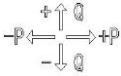
يوم : Remain days

ك.و.س : Measurement

باقى : Remain credit

: Relay closed

: Relay open

: Indicate current quadrant

L1 L2 L3

: Indicate voltage, current, reverse

Item No.	Item name	Remark
01	Current remaining energy	
02	Current remaining credit	
03	Current remaining days	
04	Cumulative total forward active energy	No decimal
05	Current month consumption energy	
06	Current step tariff value	
07	Current step tariff No.	
08	Maximum current limit	Single phase
09	Current month consumption credit	
10	Current step number	
11~19	Current step1~9 value	If no value, display NULL
20	Current step tariff number	
21~30	Current step tariff1~10 value	
31	Current date	
32	Current time	12 hours format A-: Morning P-:Afternoon
33	Current week day	1~7
34	Remain credit alarm limit 1	

35	Remain credit alarm limit 2	
36	Remain credit alarm limit 3	
37	Hoarding threshold	
38	Overdraft threshold not friendly	
39	Active pulse constant	
40	Reactive pulse constant	
41	Serial number of vending system	
42	Customer number high 8 digits	
43	Customer number middle 6 digits	
44	Customer number low 6 digits	
45	Meter number high 4 digits	
46	Meter number middle 8 digits	
47	Meter number low 8 digits	
48	Remain days alarm limit	
49	Cycle display interval	Unit: seconds
50	Cycle display items number	
51	Actual power on days current month	
52	Average day consumption of last month	
53	Total purchase credit	
54	Times of purchase	
55	Terminal cover open times	Unit: kWh
56~57	Latest terminal cover open date and time	If there is no event, it will display: NULL
58	Power off times	
59~60	Latest power off date and time	If there is no event, it will display: NULL

61~72	Last 1~12 month: active energy of each month	
73~108	Last1~12month: Max Demand, happen date and time	
109	Current reverse active energy	
110~113	Alarm reasons	Details please check the note1
114~116	Trip reasons	Details please check the note2
117	The left time of open cover the relay don't open	Unit: second
118	Power on times	
119~120	Latest power on date and time	
121	Unbalance limit	
124	Power Limit value	
129	Lowe voltage trip value	
130	CT ratio	
131	Phase A voltage	
132	Phase B voltage	
133	Phase C voltage	
134	Phase A current	
135	Phase B current	
136	Phase C current	
137	Current of neutral line	
144	Total active power	
145	Phase A active power	
146	Phase B active power	
147	Phase C active power	
151	Reactive power	

152	Phase A reactive power	
153	Phase B reactive power	
154	Phase C reactive power	
155	Total power factor	IndXXXXXX indicate inductive CAPXXXXXX indicate capacitive
156	Phase A power factor	
157	Phase B power factor	
158	Phase C power factor	
159	Frequency	
160	Phase disorder happen times	
161~162	The latest phase disorder happen date and time	
163	Reverse happen times	
164~165	The latest reverse happen date and time	
166	Phase loss happen times	
167~168	The latest phase loss happen date and time	
169	Over voltage happen times	
170~171	The latest over voltage happen date and time	
172	Unbalance happen times	
173~174	The latest unbalance happen date and time	
175	When the latest unbalance happen the unbalance limit	
176	When the latest unbalance happen phase A active power	

177	When the latest unbalance happen phase B active power	
178	When the latest unbalance happen phase C active power	
179	The relay trip times because of the remain credit using up	
180~181	The latest relay trip date and time because of the remain credit using up	
182	Reset happen times	
183~184	The latest reset happen date and time	
185	External battery low voltage happen times	
186~187	The latest external battery low voltage happen date and time	
188	Magnetic disturbance happen times	
189~190	The latest magnetic disturbance happen date and time	
191	Over MPL happen times	
192~193	The latest over MPL happen date and time	
194	Meter cover open times	
195~196	The latest meter cover open date and time	
197	Partially missing neutral happen times	
198~199	The last time partially missing neutral happen date and time	
200	Demand period	
201	Demand period number	
204	Software test finish time	

205	Software version	
206	Hardware version	
207	Current terminal cover status	
208	Total overdraft credit	
209	Total overdraft energy	
210	Power direction	Details please check note 3
214	Billing date	
215	Friendly start time	
216	Friendly finish time	
217	Weekend	From left to right: Saturday Friday Thursday Wednesday Tuesday Monday Sunday 0 means weekend, even no energy the relay will not open 1 means work day
218	Prepayment indication	Bit0: 1 means the meter has been open account Bit 1: 1 means the meter has been changed meter(data from change meter card) Bit 2:1 means the meter has been canceled account Bit 3: 1 means the meter has been changed
219	Friendly indication and TOU holiday indication	Bit0:1 means friendly day Bit1:1 means friendly time Bit2:1 means weekend Bit3:1 means TOU holiday

220	Offset cover steps	
221~230	Offset value cover from 0~9	
231	TOU tariff number	
232~239	TOU tariff1~8 value	
250~252	Each phase total active energy	
253	Left time when pressing the parameter setting button	
254	Protocol of optical	
255	Protocol of RS485	
256	Average power factor	
257	relay control status	Disconnect: force relay disconnect Connect: force relay connect NULL: no force control
258	Delay time of relay control card	
259	Over voltage limit	
300~307	Meter mode word 0~7	The details please check note 4.
308	Friendly day number	
309	Next friendly date	yymmdd, if no display NULL
310~339	Friendly day	
340	Actual power on days last month	
341~342	Open account card in setting date and time	
343	Open account credit	
344~352	Last 3 times purchase record	

356~357	Cancel account card in setting date and time	
358	Remain credit before in setting cancel account card	
359~360	Change meter card inserting date and time	
361	Remain credit of change meter card	
362~363	Inset card to remove open cover trip date and time	
364	Inset card to remove open cover trip operator No.	
365~366	Last 1 time remote charge date and time	
367	Last 1 time remote charge credit	
368	Parameter setting card inset times	
369-370	Last 1 time parameter setting card inset date and time	
372	Friendly time overdraft limit	
373	Load profile record interval	
374	Special radio fees	The default one is .002
375	Special radio energy limit	The default one is 45kWh
376	Special tariff 1	The default one is 0.006EGP
377	Special tariff energy 1	The default one is 10kWh
378	Special tariff 2	The default one is 0.003EGP
379	Special tariff energy 2	The default one is 10kWh
380	No consumption for one month the deduct credit	The default one is 3EGP
381	Current month deduction of special radio fees	

382	Current month deduction of special tariff 1	
383	Current month deduction of special tariff 2	
384	Total deduction of special radio fees	
385	Total deduction of special tariff 1	
386	Total deduction of special tariff 2	
387	Total No consumption for one month the deduct credit	
388	Prepayment mode consumption energy	
389	Postpaid mode consumption energy	
390	Prepayment and postpaid indication	Prepayment: PrEPAY ; Credit: POST PAY
391	Current step tariff table No.	
393~395	Reverse energy of phase A/B/C	
396	Reactive energy of quadrant 1	
397	Reactive energy of quadrant 2	
398	Reactive energy of quadrant 3	
399	Reactive energy of quadrant 4	
401	Actual max power limit switch times every day	
402~465	Max power limit switch list	Format: Hour time, MPL
466	Billing times	
467~478	Last 1~12 month total active energy	Less than 9999999.9kWh
479~490	Last 1~12 month active energy of each month	
491-502	Last 1~12 month total active energy	More than 9999999.9kWh

503~504	Active power demand starting date and time	
505~506	Reactive power demand starting date and time	
507~508	Apparent power demand starting date and time	
509~510	Current demand starting date and time	
511~513	Current month the maximum power demand and happen date and time	
514~516	Current month reverse maximum power demand and happen date and time	
517~519	Quadrant 1 reactive maximum power demand and happen date and time	
520~522	Quadrant II reactive maximum power demand and happen date and time	
523~525	Quadrant III reactive maximum power demand and happen date and time	
526~528	Quadrant IV reactive maximum power demand and happen date and time	
529~531	Apparent maximum power demand and happen date and time	
532~534	Apparent reverse maximum power demand and happen date and time	
535~537	Current month the Max current demand and happen date and time	
538	Total maximum demand kW	
539	Total maximum demand A	
540	Relay error happen times	

541-542	The latest relay error happen date and time	
543	Relay error kWh	
544	Consumption energy when missing neutral	
545	Standby offset value actual used number	
550	Current TOU tariff value	
551	Current TOU holiday total number	
552	Current TOU week table No.	
553	Current TOU daily table No.	
554	Current TOU tariff No.	
555	Current TOU time interval number	
556~569	Current TOU daily table display	
570	Current TOU week table	
571	Current TOU season number	
572-585	Details of season table	
591-598	Consumption energy of each tariff	
601-612	Last 1~12 each month power factor	
599	Q1+Q2	
600	Q3+Q4	
613	Q1+13	
614	Q2+Q4	
615	Q1+Q2+Q3+Q4	
616-618	Phase angle of A/B/C	
619	Angle between phase A voltage and phase B voltage	

620	Angle between phase A voltage and phase C voltage	
621	One phase with current without voltage happen times	
622~623	One phase with current without voltage happen date and time	
624	Calculate neutral current	
625	Totally missing neutral times	
626-627	The last time totally missing neutral happen date and time	
628~630	Old meter number before changing meter	
631	Total energy when latest open terminal cover happen	
632	Total consumption when latest power off happen	
633	Total consumption when latest power on happen	
634	Total consumption when latest open top cover happen	
635	Total consumption when latest strong magnetic alarm happen(reserve)	
636	Total consumption when latest battery alarm happen	
637	Total consumption when latest relay error happen	
638	Total consumption when latest reverse happen	
639	Total consumption when latest phases miss-order happen	

640	Total consumption when latest loss phase happen	
641	Total consumption when latest total missing neutral happen	
642	Total consumption when latest partial missing neutral happen	
643	Total consumption when latest missing voltage happen	
644	Consumption during reverse happen	
645	Consumption during Wrong phase sequence	
646	Consumption during With current without voltage happen	
647	Consumption during Phase loss	
648	Relay error credit	
649	Reactive consumption of this month	
650- 661	Last 12 month reactive energy	
663	Current offset billing deduct number	
664~673	Current offset 1-10 for billing deduction	
674	Overdraft energy	
675	Overdraft money	
676	Average PF yearly	
677	Monthly consumption limit 1	
678	Monthly consumption limit 2	
679	Over monthly consumption limit 2 special deduct value	
680	Current overdraft limit	

681	After deducting special deduct value remain credit	
682	Monthly special deduct value activate limit	
683	Buzzer silent start time	
684	Buzzer silent end time	
832	RTC indication	8025T or 6543GT
881	Prepay mode and post pay mode	
882	Full screen	
883	Remain time after FAC without linking	
884	If FAC is linking	
885	starting date of current tariff	
886	Display LAB status	Without lab test card display out lab Insert lab test card display LAB XXXX
887	Active power demand of phase A	
888	Active power demand of phase B	
889	Active power demand of phase C	

Note 1: Alarm reasons:

Screen 1 display item: 110

Bit	Meaning	Binary code display From left to right
Bit 7	Magnetic disturbance	10000000
Bit 6	Over voltage	01000000
Bit 5	missing natural/part missing neutral	00100000

Bit 4	Monthly overdraft	00010000
Bit 3	Monthly consumption over limit	00001000
Bit 2	Phase disorder	00000100
Bit 1	Open terminal cover when power off	00000010
Bit 0	Open terminal cover when power on	00000001

Screen 2 display item: 111

Bit	Meaning	Binary code display From left to right
Bit 7	Reserved	10000000
Bit 6	There is current no voltage of one phase	01000000
Bit 5	Relay fault	00100000
Bit 4	External battery low voltage	00010000
Bit 3	Reserved	00001000
Bit 2	Low voltage	00000100
Bit 1	Reverse	00000010
Bit 0	Phase loss	00000001

Screen 3 display item: 112

Bit	Meaning	Binary code display From left to right
Bit 7	Memory card initialize fault	10000000
Bit 6	Memory card fault	01000000
Bit 5	MPPUL	00100000
Bit 4	Remain credit less than alarm limit 3	00010000

Bit 3	Remain credit less than alarm limit 2	00001000
Bit 2	Remain credit less than alarm limit 1	00000100
Bit 1	Clock start from power on	00000010
Bit 0	RTC fault more than 10 times	00000001

Screen 4 display item: 113

Bit	Meaning	Binary code display From left to right
Bit 7	Remain credit used up and reach to the overdraft limit not friendly	10000000
Bit 6	Remain days less than remain days limit	01000000
Bit 5	Average current of 3phase over MPL	00100000
Bit 4	Phase A over MPL	00010000
Bit 3	Phase B over MPL	00001000
Bit 2	Phase C over MPL	00000100
Bit 1	Open meter cover when power off	00000010
Bit 0	Open meter cover when power on	00000001

Note 2: Replay open reasons:

Screen 1 display item: 114

Bit	Meaning	Binary code display From left to right
Bit 7	Low voltage	10000000
Bit 6	Over voltage	01000000

Bit 5	Reserved	00100000
Bit 4	Reserved	00010000
Bit 3	Reserved	00001000
Bit 2	Phase disorder	00000100
Bit 1	Open terminal cover when power off	00000010
Bit 0	Open terminal cover when power on	00000001

Screen 2 display item: 115

Bit	Meaning	Binary code display From left to right
Bit 7	Remain credit used up and reach to overdraft limit	10000000
Bit 6	Relay open when remain credit is 0 except friendly	01000000
Bit 5	average current over limit	00100000
Bit 4	Phase A over MPL	00010000
Bit 3	Phase B over MPL	00001000
Bit 2	Phase C over MPL	00000100
Bit 1	Reverse	00000010
Bit 0	Phase loss	00000001

Screen 3 display item: 116

Bit	Meaning	Binary code display From left to right
Bit 7	Reserved	10000000
Bit 6	Remain credit less than alarm limit 3	01000000
Bit 5	Remain credit less than alarm limit 2	00100000

Bit 4	Remain credit less than alarm limit 1	00010000
Bit 3	Reserved	00001000
Bit 2	Reserved	00000100
Bit 1	Open meter cover when power off	00000010
Bit 0	Open meter cover when power on	00000001

Note 3: Power direction

Bit	Meaning	Binary code display From left to right
Bit 7	Reactive reverse	10000000
Bit 6	Phase C reactive reverse	01000000
Bit 5	Phase B reactive reverse	00100000
Bit 4	Phase A reactive reverse	00010000
Bit 3	Active reverse	00001000
Bit 2	Phase C active reverse	00000100
Bit 1	Phase B active reverse	00000010
Bit 0	Phase A active reverse	00000001

Note 4: Mode word 0:

Bit	Meaning	Binary code display From left to right
Bit 7	TOU tariff	10000000
Bit 6	Step tariff(fixed 1)	01000000
Bit 5	Deduct unit: 0: 1kWh 1: 0.01kWh	00100000
Bit 4	1:prepayment 0:postpayment	00010000

Bit 3	Reserved	00001000
Bit 2	Reserved	00000100
Bit 1	Reserved	00000010
Bit 0	Reserved	00000001

Mode word 1: reserved

Mode word 2:

Bit	Meaning	Binary code display From left to right
Bit 7	Control whether relay open when remain credit less than alarm limit 1	10000000
Bit 6	Control whether relay open when remain credit less than alarm limit 2	01000000
Bit 5	Control whether the relay open when open terminal cover before open account	00100000
Bit 4	Control the event record 1 time or many times	1: record 1time 0:record many times
Bit 3	Control whether the relay open when weekend	1: relay not open 0:relay open
Bit 2	Control whether the relay open when friendly time	1: relay not open 0:relay open
Bit 1	Control whether the relay open when friendly day	1: relay not open 0:relay open
Bit 0	Control whether the relay close when inset legal card and the overdraft energy not reach to overdraft limit in the friendly time	1.relay close 2. Relay don't close

Mode word 3:

Bit	Meaning	Binary code display From left to right
Bit 7	Control whether relay open when reverse happen	10000000
Bit 6	Control whether relay open when phase loss happen	01000000
Bit 5	Control whether relay open when remain credit is 0 except friendly time	00100000
Bit 4	Control whether relay open when phase disorder	00010000
Bit 3	Control whether relay open when power off	00001000
Bit 2	Control whether relay open when meter cover open after open account	00000100
Bit 1	Control whether relay open when remain credit less than alarm limit 3	00000010
Bit 0	Control whether relay open when terminal cover open after open account	00000001

Mode word 4: Reserved

Mode word 5:

Bit	Meaning	Binary code display From left to right
Bit 7	Overdraft buzzer alarm	10000000
Bit 6	Monthly consumption over limit alarm	01000000
Bit 5	Overdraft mode (0: overdraft energy 1: overdraft hour)	00100000
Bit 4	When overdraft remain credit with minus or not	00010000

Bit 3	Activate friendly limit	00001000
Bit 2	Activate hoarding limit	00000100
Bit 1	Activate current limit table	00000010
Bit 0	Activate power limit function	00000001

Mode word 6:

Bit	Meaning	Binary code display From left to right
Bit 7	Control whether switch to postpaid mode	10000000
Bit 6	Control whether add 'check-xx' in cycle display	01000000
Bit 5	Control LCD switch off when there is no action	00100000
Bit 4	Control time display format:12 hours	00010000
Bit 3	Control whether display the error information according to check	00001000
Bit 2	Control whether display by press button when power off	00000100
Bit 1	Control display energy with 2 decimal	00000010
Bit 0	Reserved	00000001

Mode word 7:

Bit	Meaning	Binary code display From left to right
Bit 7	Control the protocol of RS485 is not standard 1107.	10000000
Bit 6	Buzzer: 1: no resource(pulse) drive	01000000

	0: resource(DC) drive	
Bit 5	Control if using 6543GT internal RTC	00100000
Bit 4	Missing neutral detection method 1: compare calculate and measure 0: hardware detection	00010000
Bit 3	Relay error detection method: 1: hardware 2: current when disconnect	00001000
Bit 2	Current demand: 1: total current 0: average current	00000100
Bit 1	Over MPL detection method: 1:average of 3phase 0: each phase	00000010
Bit 0	Control whether buzzer alarm	00000001

Error display type :

No .	Display	Meaning
-01	tEr-OPEN	Terminal cover open during power on;
-02	tEr-OPEN	Terminal cover open during power off
-03	E-SEqUEN	Phase disorder
-04	toP_OPEN	Meter cover open during power on
-05	toP_OPEN	Terminal cover open during power off
-06	MISS-NE	Missing neutral

-07	OUEr-U	Over voltage
-08	MAgNEt	Magnetic disturbance ;
-09	droP-U-I	Phase loss of phase A
-10	rEUErSE	Current revers
-11	E-bAttEr	Battery low voltage
-12	E-rELAY	Relay fault
-13	MISS-U	Only current no voltage
-14	OVEr-p	Over power limit
-15	L-LEU1	Remain credit less than limit1
-16	L-LEU2	Remain credit less than limit2
-17	L-LEU3	Remain credit less than limit3
-18	rELAY-F	Force control relay
-19	Time value	Force control relay time
-20	UNdEr-U	Low voltage
-21	OUErxxxx	Monthly consumption over limit
-22	dEbtor	Monthly overdraft
-23	Drop-U-2	Loss phase of phase B
-24	Drop-U-3	Loss phase of phase C

5.3 **LED indications**

From left to right:

Active pulse: red

Reactive pulse: red

Low credit indicate: double color green and red

Alarm indicate: red

When there is control cable input on the auxiliary terminal, there is no alarm LED.

6. Prepayment

When inset card, LCD display CArd_XX, XX is card type. If the meter cannot identify the card, XX will display 00.

If the card accepted by the meter, the buzzer will prompt 1 time and LCD will display الكارت مقبول .

If the card rejected by the meter, the buzzer will prompt 3 times and LDC will display الكارت غير , and display Err-XXYY, XX is card type, YY is fault code. If the customer credit card inset again without new credit the meter will prompt 'USED'

Error Code	Description
01	Data cannot be read
02~30	Data frame verification error
31~35	32K read card data frame verification error
36	32kB card index error
37	32kB card password protect area CRC check error
38	32kB card set area CRC check error
39	32KB card password check error
40	Card type error
41	The meter type data is not 3 phase in the card
42	Cannot support prepay fee
43	Vending system serial number error
44	Purchase credit too more over 999999.99
45	Customer No error
46	Meter No error (this card already turn back other customers information)

47	Meter No error (when need judge the meter number in the card or the meter number in the charge card or mend card is not right)
48	Preset energy of each phase over 10000000kWh
49	Credit is more than 999999.99
50	Initialization card return data error
51~55	Pull out card too early before the meter finishing reading
57	Pull out card too early, when insetting 32K read card
58	Pull out card too early, when copy the data from meter to card
59	Pull out card too early, when copy the data from card to meter
60~63	CPU Card approved error
64	32kB card user No error
70	The lead seal button is not be pressed
71	Status error, the meter cannot accept purchase card before open account
72	The meter is already cancel account
73	Relay check of last time is not finish
74	Address back to card over the Max address(check and reading card)
75	The meter has been open account cannot accept setting card
76	Reserved
77	Reserved
78	Data back to card is bad position(the first 64 byte of CPU Card)
79	Data back to card is bad position(over 0x3D0)

80	The value in the card add the remain credit in the meter is more than hoarding threshold or more than 999999.99
81	Purchase times error: times in the card is more than times in the meter
82	Purchase times error: times in the card is less than times in the meter
83	Terminal cover is not covered well when removing open cover alarm
84	Meter cover is not covered well when removing open cover alarm
85	Meter cover and terminal cover are not covered well when removing open cover alarm
86	The lead seal button is not pressed when copy the data from meter to card
87	The lead seal button is not pressed when copy the data from card to meter
88	Pull out card too early when copy data from meter to card and the meter number is different
89	Pull out card too early when copy data from card to meter and the meter number is different.
90	Still over MPL when removing over MPL alarm
91	Over limit time
92	Not 32KB change card(old meter is till good meter)
93	Event still exist
94	Not 32KB change card(old meter is damaged meter)
95	Not press the program button when inset change card
96	The firmware of old meter is too old
97	The change meter card already used

98	When paste the data from card to new meter, there is no old data in card
99	The data in card is not complete
9A	Pull out card too early when change meter
9B	The tariff setting card don't have suitable tariff scheme
A1	CPU card operation, pull out card too early
A2	CPU card operation, read data error
A3	CPU card operation, write data error
A4	CPU card operation, open file error
A5	CPU card operation, internal authentication error
A6	CPU card operation, external authentication error

Card Types

Initialization card (new customer card)

One initialization card should just use in the fixed corresponding meter. It can set the customer ID with the corresponding system number. Can set the remaining energy value or accumulator the energy value. The card can clear all of data as the customization selection energy or events data. The card has the special password for the data security. And can return read currently data from the meter.

Card type code: 01H

Customer credit card

Just for the customers. Customers use this card to purchase the credit and get energy in the meter running state.

One initialization card should just use in the fixed corresponding meter while with the special password for the data security. And can return read currently data from the meter.

Card type code: 02H

Reissue card

The vending department should reissue the new card for the customers when customer's card had losses or broken.

Card type code: 04H

In the base of the all operation successes, when the purchased times of CPU Card is greater 1 than meters, the credit can save in the meters and return data from meter to cards. If the purchased times is same between meters and card, just return currently meter data to CPU Card. If that is in other conditions, the meter will alarm as – purchased time error.

Account canceling card

Use for cancel customer account that can read currently remain energy and clear the energy value via push the programming button.

One initialization card should just use in the fixed corresponding meter while with the special password for the data security. And can return read currently data from the meter.

Card type code: 11H

Parameters card

This card can set the parameters for the meters.

This card can use in any meters with the communal password. It can return meter No and meter status.

Card type code: 10H

Check and reading card

This card can read parameters and energy data from meter.

This card can check the relay disconnect. When inset the card the relay will action, after 10 seconds the relay will action.

Card type code: 13H

Change meter card (the old meter is good meter)

The card is used to copy data from meter A to meter B. This operation need press the lead seal button.

Card type code: 09H

Change meter card (the old meter is damaged meter)

Using VTC communication read the memory card of meter A, then write to a change meter card, then inset the card in meter B.

Card type code: 19H

Preset card

This card is used to preset credit in the meter before open account

Card type code: 05H

32kB check and reading card

This card can read 250pcs, reading data including: meter No. tariff No. remain credit, meter status(alarm reason), total consumption energy, each month consumption energy of last month's 12 month, each month average power factor of last 12 months

This card no need to check the user No.

You can choose whether check the relay when inset card.

7. Communication

Optical communication

Optical communication for local data reading and writing

Baud rate: 300bps

Format: start bit 1bit, data bit 7 bits, check bit 1bit, stop bit 1bit

7.1 Relay control Signal

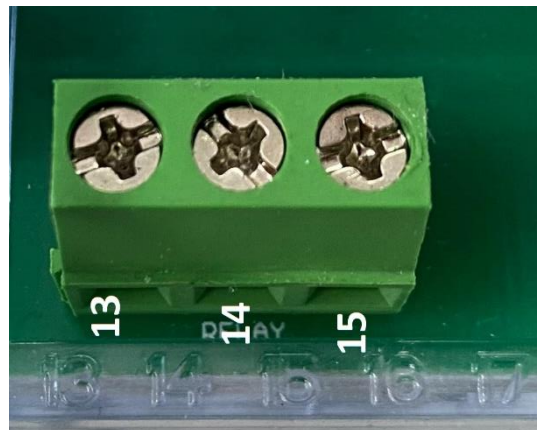
Meter gives output signals if the relay connected or disconnected to control the MCB outside the meter in order to control the main load

Three output terminals (13-14-15) defined as:

Point 13: output signal the relay is disconnected.

Point 14: common.

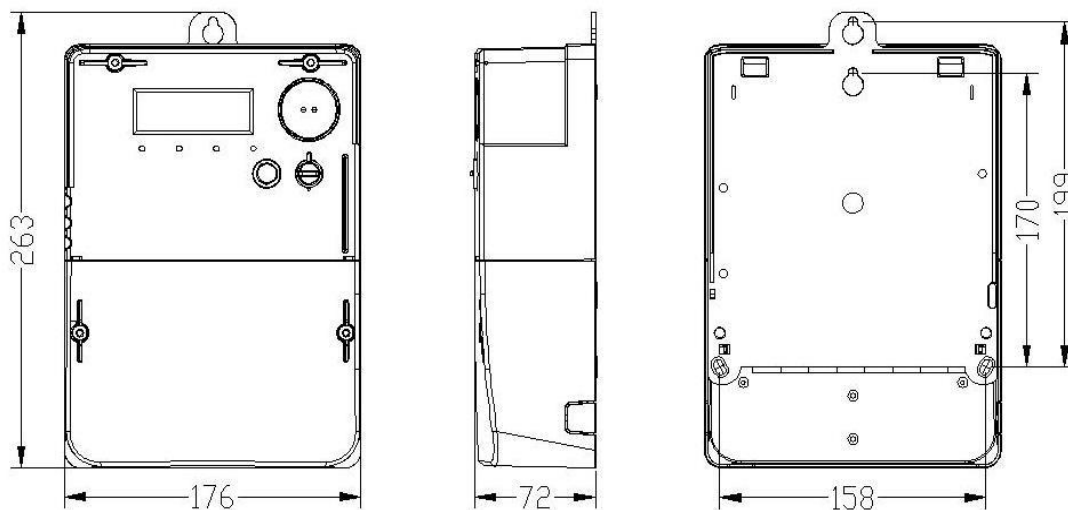
Point 15: output signal the relay is connected.



8. Installation and connection

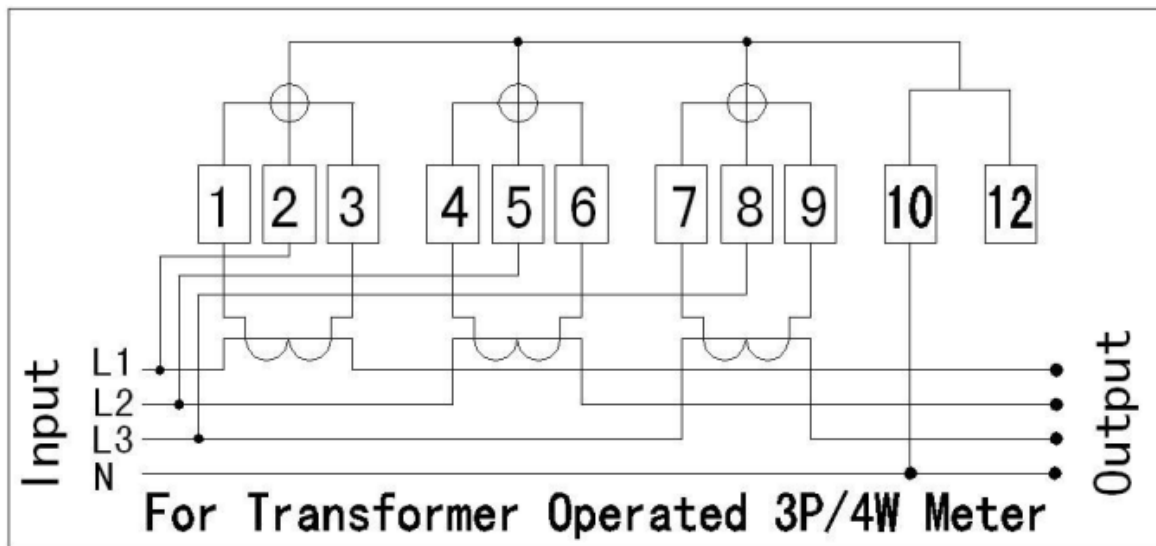
8.1 Meter installation

Meter install dimension as below, there is a hook at the upper of meter back, using M6 hook screw for fixing. And there are 2 mounting holes at the lower of meter, using M4 screw or ST4.2 x12 self-trap screws.



OUTLINE DIMENSION DRAWING

8.2 Wire connection diagram as following



Wire connection diagram

9 Transport and storage

★Meter transport should be without severe vibration and shock. Its storage should be consistent with transport packaging JB/T9329-1999 and GB/T15464-1995.

★Product should be in the original packaging box, the environment should be kept clean and the ambient temperature does not exceed $-20\text{ }^{\circ}\text{C} \sim +65\text{ }^{\circ}\text{C}$, relative humidity less than 85%, non-corrosive gases in the air and the mold.

★If product stored in the warehouse, it should be placed on shelves with original packing, and the stacking height should be not exceeding 5 layers.

★Product should not be stored after open packing.